

Homework 2

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Part 1

Claim from homework assignment: Let (a_n) and (b_n) be sequences of nonnegative real numbers. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c \in (0, \infty),$$

then the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are either both convergent or both divergent.

Proof. Since $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\frac{c}{2} < \frac{a_n}{b_n} < 2c.$$

This implies

$$\frac{c}{2}b_n < a_n < 2cb_n \quad \text{for all } n \geq N.$$

1. If $\sum b_n$ converges, then by the Comparison Test, since $0 \leq a_n \leq 2cb_n$ for large n , it follows that $\sum a_n$ also converges.

2. If $\sum b_n$ diverges, then from $a_n \geq \frac{c}{2}b_n$ for sufficiently large n , it follows that $\sum a_n$ must also diverge. Otherwise, we would have a contradiction to the divergence of b_n (a nonnegative series cannot be bounded above by a convergent one if it were to diverge).

Hence, $\sum a_n$ and $\sum b_n$ share the same nature: they either both converge or both diverge. $\square$

Part 2

Claim from homework assignment: Given a sequence (a_n) of nonnegative real numbers, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ converges.

Proof. First, observe that for all n :

$$0 \leq \frac{a_n}{1+a_n} \leq a_n.$$

If $\sum a_n$ converges, then $a_n \rightarrow 0$. Hence,

$$\lim_{n \rightarrow \infty} \frac{\frac{a_n}{1+a_n}}{a_n} = \lim_{n \rightarrow \infty} \frac{1}{1+a_n} = 1.$$

By the Limit Comparison Test (as established in Part 1), since the limit of the ratio is a positive finite number, it follows that $\sum \frac{a_n}{1+a_n}$ converges if and only if $\sum a_n$ converges.

Conversely, if $\sum \frac{a_n}{1+a_n}$ converges, then the terms $\frac{a_n}{1+a_n} \rightarrow 0$. This forces $a_n \rightarrow 0$. Applying the same limit comparison argument again, we deduce $\sum a_n$ converges.

Thus, $\sum a_n$ converges if and only if $\sum \frac{a_n}{1+a_n}$ converges. □

References

I came up with this proof and all its parts by myself. Then, I used ChatGPT o1 to convert it into a LaTeX document for readability.